TSM 099

Homework 9

Percents, models, and contextual systems

Topics 35-37 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Translate percent relationships

1. Convert (a) 0.6% to a decimal, (b) 1.25 to a percent, and (c) 7/8 to a percent.

2. A class has 32 students; 28 complete an assignment. Find the completion percentage and the number that did not complete it.

3. A sale price of $68 is 80% of the original price. Define the unknown, write an equation, and find the original price.

4. A quantity increases from 48 to 60, then decreases from 60 to 48. Find both percent changes and explain why they differ.

Successive changes and equations

5. A $200 item is discounted 15%, then taxed 6% on the sale price. Find the total and the net percent change from $200.

6. After an 8% increase, a fee is $81. Find its original amount. Explain why subtracting 8% of $81 is not the inverse calculation.

7. A number plus 30% of itself is 91. Write and solve an equation.

8. A rectangle has perimeter 46 cm, and its length is 5 cm more than its width. Find its dimensions and area; check the perimeter.

Systems and reasonableness

9. A venue sells 30 tickets. Standard tickets cost $12; discounted tickets cost $8. Revenue is $304. Define variables, solve a system, and check both totals.

10. Mix a 10% solution and a 30% solution to obtain 10 liters of a 22% solution. Assume volumes add. Define variables, write a total-volume equation and a solute equation, then solve.

11. Two rental plans cost A = 18 + 3h and B = 6 + 5h dollars for h hours. Find when the costs are equal. Use a check at h = 2 and h = 10 to decide which is cheaper on each side of that point.

12. A model gives -3 as the number of tickets to buy. Explain why an algebraic solution may not be a valid answer to a word problem, and name two checks to perform.